Human Stress Analysis Between Normal and ADHD Patient by Using Neural Networks Based on EEG Signals

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***Abstract*—Electro Encephalogram (EEG) is used to read electrical signals from all portions of the brain by attaching electrodes on the scalp. These signals are recorded and analyzed on a patient with a form of neurodevelopmental disorder called Attention-Deficit/Hyperactivity Disorder (ADHD) and a control patient. We compare brain activity of both patients on their resting states, emotional states and the focus of this study, their reaction to stressful stimuli.**

Keywords—Electro Encephalogram (EEG), Neural Networks, Attention-Deficit/Hyperactivity (ADHD), Convolutional Neural Network (CNN), Multi-Layer Perceptron (MLP)

I. Introduction

The purpose of this project is compare the differences in brain signal activity between an experimental subject and a control subject under stress-test condition. Specifically, the experimental subject is a student who has Attention-Deficit/Hyperactivity Disorder (ADHD) while the control subject is a normal student with no known mental health condition. The brain signal will be recorded using an EEG machine and the EEG signals then will be analyzed using a neural network. Proper signal processing steps will be applied throughout the project such as pre-processing, extracting features and classifying the signals.

II. Literature Review

A. A Deep Learning Framework for Identifying Children with ADHD using an EEG-Based Brain Network

Chen H et al. (2019) conducted a study on identifying children with ADHD using EEG-based brain networks. Their main objective is to develop a deep learning algorithm using convolutional neural network (CNN) that can detect ADHD symptoms in these children. A ten-minute resting state EEG was the proposed protocol with conditions like the subjects had enough sleep and avoided any stimulants the night prior. The pre-processing technique used is an automatic algorithm developed by P.J. Durka et al. (2003) while mutual information (MI) method is used to extract features. Lastly, the data is classified using CNN specifically the connectivity matrix. The developed model produced good results with an accuracy of 94.67% when detecting ADHD symptoms.

B. Deep Learning Bassed on Even-Related EEG Differentiates Children with ADHD from Healthy Controls

Another research was conducted by Vahid A et al. (2019) aiming to differentiate children with ADHD from healthy control children. The data collection protocol includes requiring the participants press a button after the 12 seconds mark according to their estimation or intuition. The task was divided into three blocks, and the participants were given one minute break time between blocks. The collected data then will be filtered using band-pass filtering between 0.5 and 20 Hz and cleaned by removing pulse and eye movement artifacts using independent component analysis (ICA) and infomax algorithm. No specific feature extraction method was mentioned but they chose signals with amplitude between 200 to -200 in 200ms interval. A deep learning classifier, specifically EEGNet is used as the classifier to classify the data. The researchers reported up to 83% accuracy when distinguishing ADHD children from healthy control children.

C. Deep Learning Convolutional Neural Networks Discriminate Adult ADHD from Healthy Individuals on the Basis of Event-Related Spectral EEG.

An identical study also has been conducted by Debreuil-Vall et al. (2020) but on adults instead of children. The objective of the study is to develop a deep learning convolutional neural network model that can identify adult ADHD patients from healthy individuals. The protocol proposed is of Eriksen-Flanker Task (EFT) where the participants have to focus and respond to directions of an arrow surrounded by small distractions. Also, brain signals were recorded when the participants were in the resting state for comparison purposes. The ICA was utilized in the pre-processing phase to remove artifacts and the signals were filtered from 1 to 20 Hz only to exclude non-neural physiological activity. The Wavelet transform was chosen as the feature extraction method to create event-related potential

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(ERP) spectrograms. Finally, the data were classified by using Tensorflow, a CNN algorithm that contains a relatively simple four-layer convolutional network. This combination of signal processing was able to achieve an accuracy of 88% in attempt to identify adult ADHD patients among healthy controls.

III. Experimental Protocol

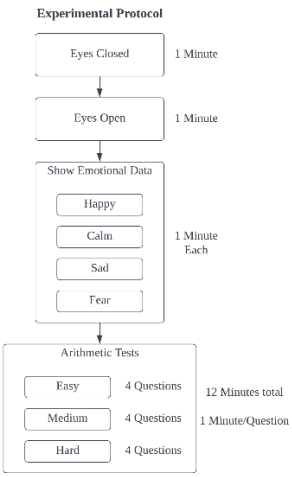


Fig. 1: Experimental Protocol

The collection of patient data follows the above protocol, the metrics used in this experiment are

1. Resting State
2. Emotional Data
3. Arithmetic Data

The resting state includes "Eyes Closed" for one minute and "Eyes Open" for one minute. The subject will be asked to perform these actions without any motor actions (no moving), this data will be used to asses the subject's baseline readings. Next the subject will be shown 4 videos of 4 emotions (Figure 1), this is to get EEG readings on how the subjects respond to emotional stimulus, which then can be used to train a MLP model to classify what the subjects are feeling when performing "Eyes Closed", "Eyes Open" and arithmetic test. The arithmetic test is to gauge how the subjects respond to problem solving or memory testing, the questions are split into 3 sections of increasing difficulty (Figure 1).

IV. Raw Signal Analysis

Before proceeding with the experiment and following the methodology, we have to briefly analyze the raw EEG data (already filtered using Ellipord filter) of both control and ADHD patient to see if there are any trends and patterns with the brain activities.

A. *Analyzing Raw EEG for Eyes Closed & Eyes Open*

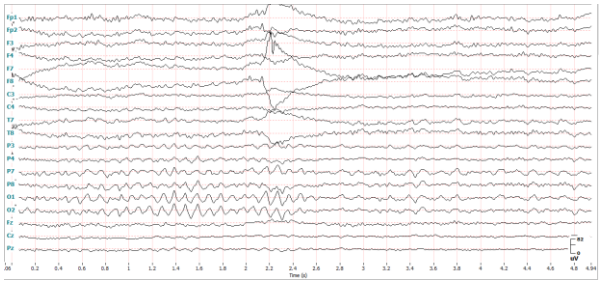


Fig. 2.1: Control Patient (Eyes Closed)

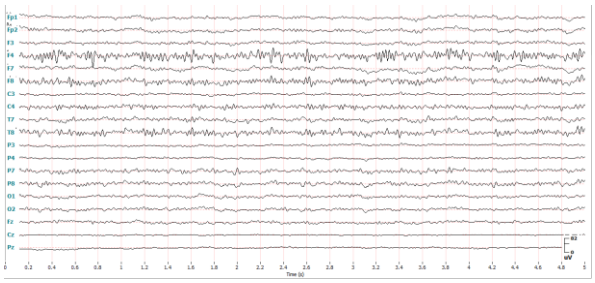


Fig. 2.2: ADHD Patient (Eyes Closed)

For eyes closed, it should trigger the occipital lobe the most due to the subject's vision impaired. The frequencies of channel O1 and time 1-12 on both control and ADHD patient belong to Alpha and Beta band waves respectively.

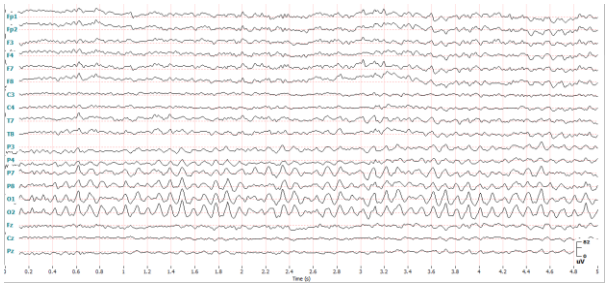


Fig. 3.1: Control Patient (Eyes Open)

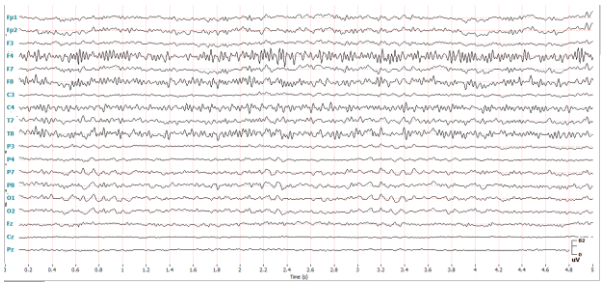


Fig. 3.2: ADHD Patient (Eyes Open)

The same can be said for eyes open as both control and ADHD patient's frequencies belong to Alpha and Beta band waves respectively. From this we can infer that the ADHD patient's brain activity is already busier than the control subject, which should probably translate in the classification section below with ADHD patients having myriad of emotions compared to control patient.

B. *Analyzing Raww EEG for Arithmetic Test*

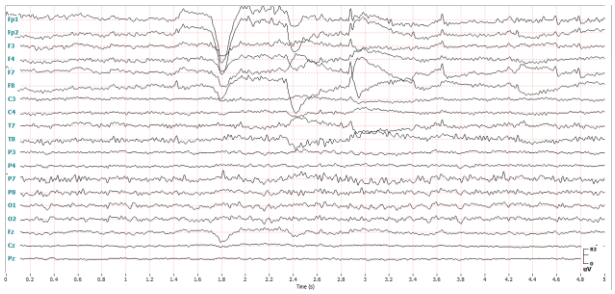


Fig. 4.1: Control Patient (Arithmetic Test)

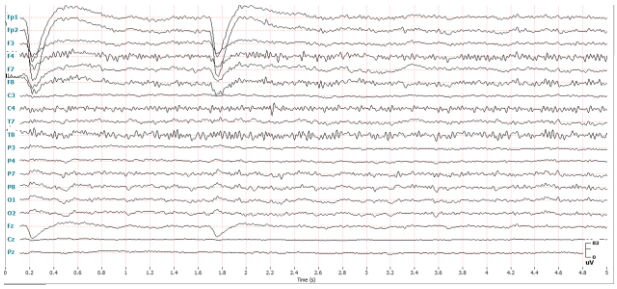


Fig. 4.2: ADHD Patient (Arithmetic Test)

The same is done with arithmetic test EEG signals, except the channel of interest is F4, which represents the frontal lobe responsible for problem solving. The frequencies of both the control and ADHD patients belong to Beta and Beta respectively. However, a thing to note here is that the ADHD patient has a higher frequency of 20 Hz while the control 15 Hz. Both are engaged, thus it is not enough information to determine anything, for this we will also analyze the temporal lobe which is responsible for memory (in this case channel T8). From the same figure (Figure 4.1 & Figure 4.2), the frequencies are Beta and Alpha. Which means that the ADHD patients are not engaging their memory functions as much as the control patient. There is bias in this finding as the intelligence of these two patients have not been accounted for, thus it is not a suitable generalization of the ADHD population.

V. METHODOLOGY

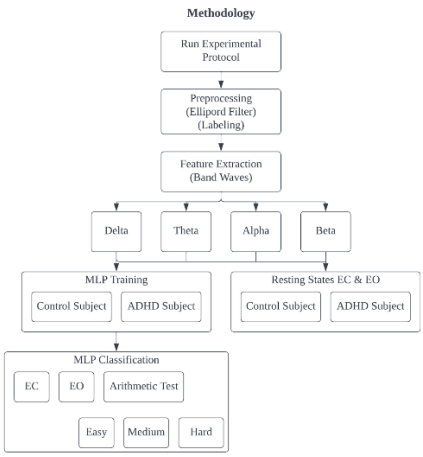
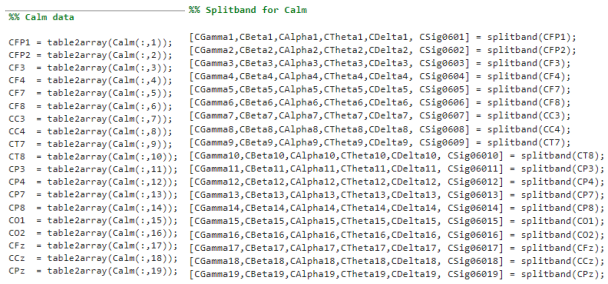


Fig. 5: Methodology

After following the experimental protocol to acquire data, said data is pre-processed on site using BrainMarker EEG software using the Ellipord filter, the Ellipord filter removes noise from the signal which originates from surroundings and electrical interferences which clears the signal for use in later stages. Using MATLAB, the individual electrode signals are split into band waves of Alpha, Beta, Delta, Theta and Gamma (Figure 6)



```
%% Calm data                                                    %% Splitband for Calm

CFP1 = table2array(Calm(:,1));   [CGamma1,CBeta1,CAlpha1,CTheta1,CDelta1, CSig0601] = splitband(CFP1);
CFP2 = table2array(Calm(:,2));   [CGamma2,CBeta2,CAlpha2,CTheta2,CDelta2, CSig0602] = splitband(CFP2);
CF3  = table2array(Calm(:,3));   [CGamma3,CBeta3,CAlpha3,CTheta3,CDelta3, CSig0603] = splitband(CF3);
CF4  = table2array(Calm(:,4));   [CGamma4,CBeta4,CAlpha4,CTheta4,CDelta4, CSig0604] = splitband(CF4);
CF7  = table2array(Calm(:,5));   [CGamma5,CBeta5,CAlpha5,CTheta5,CDelta5, CSig0605] = splitband(CF7);
CF8  = table2array(Calm(:,6));   [CGamma6,CBeta6,CAlpha6,CTheta6,CDelta6, CSig0606] = splitband(CF8);
CC3  = table2array(Calm(:,7));   [CGamma7,CBeta7,CAlpha7,CTheta7,CDelta7, CSig0607] = splitband(CC3);
CC4  = table2array(Calm(:,8));   [CGamma8,CBeta8,CAlpha8,CTheta8,CDelta8, CSig0608] = splitband(CC4);
CT7  = table2array(Calm(:,9));   [CGamma9,CBeta9,CAlpha9,CTheta9,CDelta9, CSig0609] = splitband(CT7);
CT8  = table2array(Calm(:,10));  [CGamma10,CBeta10,CAlpha10,CTheta10,CDelta10, CSig06010] = splitband(CT8);
CP3  = table2array(Calm(:,11));  [CGamma11,CBeta11,CAlpha11,CTheta11,CDelta11, CSig06011] = splitband(CP3);
CP4  = table2array(Calm(:,12));  [CGamma12,CBeta12,CAlpha12,CTheta12,CDelta12, CSig06012] = splitband(CP4);
CP7  = table2array(Calm(:,13));  [CGamma13,CBeta13,CAlpha13,CTheta13,CDelta13, CSig06013] = splitband(CP7);
CP8  = table2array(Calm(:,14));  [CGamma14,CBeta14,CAlpha14,CTheta14,CDelta14, CSig06014] = splitband(CP8);
CO1  = table2array(Calm(:,15));  [CGamma15,CBeta15,CAlpha15,CTheta15,CDelta15, CSig06015] = splitband(CO1);
CO2  = table2array(Calm(:,16));  [CGamma16,CBeta16,CAlpha16,CTheta16,CDelta16, CSig06016] = splitband(CO2);
CFz  = table2array(Calm(:,17));  [CGamma17,CBeta17,CAlpha17,CTheta17,CDelta17, CSig06017] = splitband(CFz);
CCz  = table2array(Calm(:,18));  [CGamma18,CBeta18,CAlpha18,CTheta18,CDelta18, CSig06018] = splitband(CCz);
CPz  = table2array(Calm(:,19));  [CGamma19,CBeta19,CAlpha19,CTheta19,CDelta19, CSig06019] = splitband(CPz);
```

Fig. 6: Code for Splitting into Band waves

Using MFCC feature extraction technique (to isolate all the band waves for that channel) the individual electrode channels are then compiled into their respective band waves, for e.g., all Alpha waves for all relevant electrode channels (Figure 7).

Fig. 7: Code for Compiling Band Waves for all relevant channels

These compiled band waves are the features that are used in a Multilayer Perceptron (MLP) machine learning model (Figure 8).

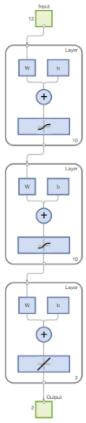


Fig. 7: MLP Network Topography

The MLP used for this experiment consists of:

1. 12 Nodes Input Layer
2. 2x10 Nodes Hidden Layer (Tanh Activation Function)
3. 2 Nodes Output Layer (Linear Activation Function)

Using Russel's model of using valence and arousal, the data is labeled as follows (a) Happy: 1,1 (b) Calm: 1,-1 and (c) Fear: -1,1 and (d) Sad: -1,-1. Eyes closed, eyes open and arithmetic test data are labeled 0,0 as the initial value.

VI. RESULTS

Following the methodology above, the results for resting state (Eyes Open & Eyes Closed) and arithmetic test can be categorized into Alpha, Beta, Delta and Theta band waves, high beta or gamma will not be utilized as there is a high chance that it is noise/artifact that slipped through the filter. All the MLP models trained for all band waves have an accuracy of at least 99% (how accurate each model is when predicting emotion based on Russel's model using test portion of dataset). The results are compiled below along with their

respective graphs (the table denotes the most emotion felt for each band wave, it does not represent the ration of the emotions).

TABLE I. CONTROL PATIENT RESULTS

Control Patient	**Tests**		
	Eyes Closed	***Eyes Open***	***Arithmetic***
Alpha	Calm	Calm	Calm
Beta	Calm	Calm	Calm
Delta	Calm	Calm	Calm
Theta	Happy	Sad	Happy
Resulting Emotion	Calm	Calm	Calm

TABLE II. ADHD PATIENT RESULTS

Control Patient	**Tests**		
	Eyes Closed	***Eyes Open***	***Arithmetic***
Alpha	Happy	Happy	Happy
Beta	Sad	Fear	Calm
Delta	Happy	Calm	Fear
Theta	Sad	Sad	Sad
Resulting Emotion	Happy/Sad	?	?

Below are the resulting valence and arousal graphs for each test, the compilation of each graph can be seen in the table above.

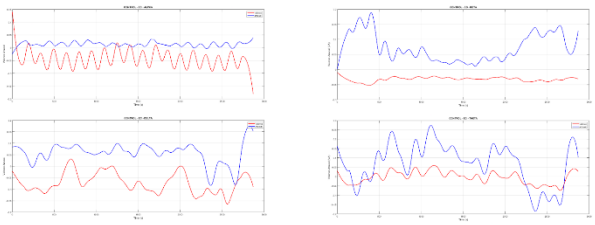


Fig. 8: Eyes Closed Control Patient

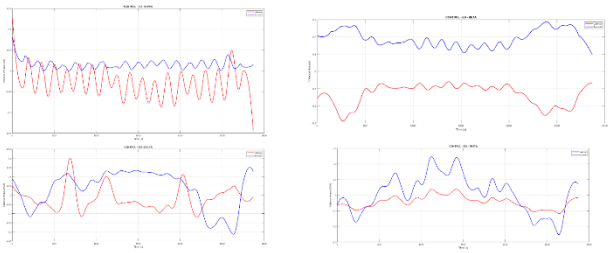


Fig. 9: Eyes Open Control Patient

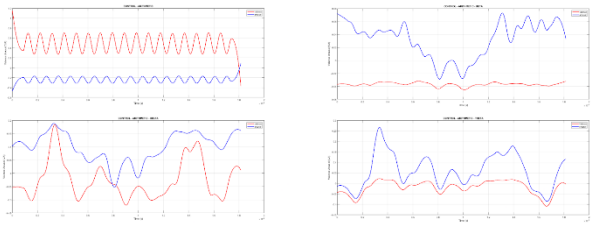


Fig. 10: Arithmetic Test Control Patient

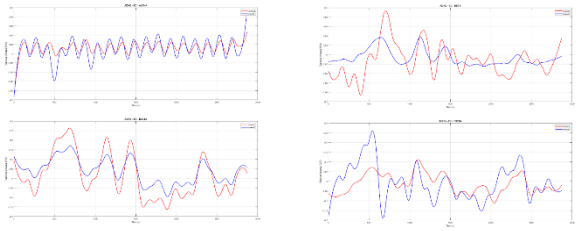


Fig. 11: Eyes Closed ADHD Patient

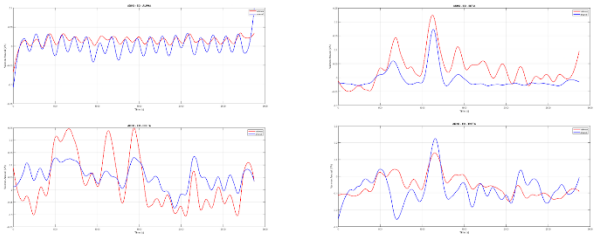


Fig. 12: Eyes Open ADHD Patient

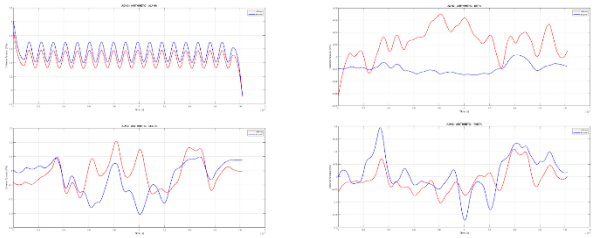


Fig. 13: Arithmetic Test ADHD Patient

VII. CONCLUSION

From signal processing we see that the brain activity of the ADHD patient is very busy, the signals are not easily interpreted(the band wave of the signals wildly varies with ADHD patient). This trend can be seen when we classify emotions of both control and ADHD patients. The emotions of the control patient can be easily identified (Table 1) where most activities the patient is calm, whereas for ADHD patients, the emotions change even within the same activity in a very short time window. This might be a potential indicator of at least identifying neurodevelopmental disorders and at best an indicator of identifying those with ADHD. ADHD patients are known to be emotionless and table 1 reflects this consensus. However the sample size is not large enough to generalize the characteristics and brain activity of all ADHD population, this can be taken as a potential trend in reinforcing other findings regarding this domain.

ACKNOWLEDGMENT *(Heading 5)*

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